A brief overview of Deep Unfolded Fractional Programming-Based Beamforming in RIS-Aided MISO Systems

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I. System Model

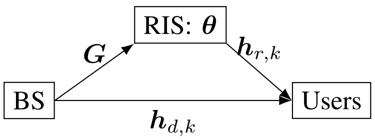

Fig. 1. A base concept of the system.

The model is described by a downlink MISO system. It consists of three main elements: the Base Station (BS), the users, and the Reconfigurable Intelligent Surface (RIS). The system has the following parameters:

- BS has M antennas.
- There are K users to be serviced.
- RIS has N elements, which causes a phase shift of $\boldsymbol{\theta} \in \mathbb{C}^N$.
- $\boldsymbol{h}_{d,k} \in \mathbb{C}^M$ is the channel matrix between the BS and the user k.
- $\boldsymbol{G} \in \mathbb{C}^{N\text{x}M}$ is the channel matrix between the BS and the RIS.
- $\boldsymbol{h}_{r,k} \in \mathbb{C}^N$ is the channel matrix between the BS and the user k.

Therefore the user k receives the signal:

$$y_k = \left(\boldsymbol{h}_{d,k}^{\mathrm{H}} + \boldsymbol{h}_{r,k}^{\mathrm{H}} \operatorname{diag}(\boldsymbol{\theta})\boldsymbol{G}\right) \sum_{i=1}^{K} \boldsymbol{v}_i s_i + n_k \tag{1}$$

from the BS. Where:

- s_i is data symbol for user i
- n_k add. white Guassian noise. $n_k \sim \mathcal{CN}(0, \sigma^2)$
- $\sum_{i=1}^{K} \boldsymbol{v}_i s_i = \boldsymbol{x}$ the signal send by the the BS.
- $\boldsymbol{v}_i \in \mathbb{C}^{Mx1}$ is the beamforming vector from the BS to the user i.
- $\boldsymbol{h}_{d,k}^{\mathrm{H}}$ is direct path between the BS and user k.
- $\boldsymbol{h}_{r,k}^{\mathrm{H}} \operatorname{diag}(\boldsymbol{\theta})\boldsymbol{G}$ is the path from the BS to RIS, application of the phase shift and then the path form the RIS to the user k.
- Summary: $y_k = (\boldsymbol{h}_{dir}^{\mathrm{H}} + \boldsymbol{h}_{indir}^{\mathrm{H}})\boldsymbol{x} + n_k$

The SINR can be calculated as:

$$\alpha_k = \frac{\left|\left(\boldsymbol{h}_{d,k}^{\mathrm{H}} + \boldsymbol{h}_{r,k}^{\mathrm{H}} \operatorname{diag}(\boldsymbol{\theta})\boldsymbol{G}\boldsymbol{v}_k\right)\right|^2}{\sum_{i=1,i\neq k}^{K} \left|\left(\boldsymbol{h}_{d,k}^{\mathrm{H}} + \boldsymbol{h}_{r,k}^{\mathrm{H}} \operatorname{diag}(\boldsymbol{\theta})\boldsymbol{G}\boldsymbol{v}_i\right)\right|^2 + \sigma_0^2} \tag{2}$$

which is simply the expected signal divided by the sum of the noise and all other signals sent.

The objective is to maximize the weighted sum rate (WSR):

$$(\mathcal{P}_1): \quad \max_{\mathbf{V},\boldsymbol{\theta}} \quad \sum_{k=1}^{K} \omega_k \log(1 + \alpha_k) \tag{3}$$

$$\text{s.t.} \quad |\theta_n| = 1, \ \forall n, \tag{4}$$

$$\sum_{k=1}^{K} \|\boldsymbol{v}_k\|^2 \leq P_{\max} \tag{5}$$

The system allows changing the beamforming vectors of the BS and the phase shift values of the RIS elements. $\boldsymbol{V} = [\boldsymbol{v}_1, ..., \boldsymbol{v}_K]$ and ω_k a weight coefficient for the user k.

This is a typical multi-ratio fractional programming problem, with a possibly non-convex solution space.

II. Block Coordinate Descent Method